RSE2203 Data Structures and Algorithms

Team Project (2026)

Introduction

The topic of data structures and algorithms span many domains ranging from robotics, artificial intelligence, data analytics, visualization, finance analysis, etc. It provides the fundamental building blocks for development and construction into more complex and sophisticated systems. A "complicated" application could be modularized while incorporated with (i) data structures that model and represent the topology and relationship of the various data, and (ii) efficient algorithms for their processing.

The objective of this team project is to address the problem scenario(s) that are set out in the application that you shall propose via development of the relevant data structures and algorithms.

Requirement and Scope

(1) Propose an application and define its problem statement(s)/scenario(s).

(2) Design and development of solution to the problem statement defined in the application.

(3) Solution would comprise the development of the relevant data structures and algorithms.

(4) Implementation of the core data structures and algorithms to be in C/C++ or Python. You may develop these via open-source libraries.

(5) Use of at least 2 or more GenAI tools (e.g. co-Pilot, ChatGPT, …etc) to help to generate basic source codes.

(6) Evaluate GenAI's Responses based on the following criteria:

   a. Accuracy: Is the information provided by the chatbot correct?

   b. Completeness: Does the chatbot provide a complete answer, or does it miss out on important details?

   c. Clarity: Is the chatbot's explanation clear and easy to understand?

   d. Relevance: Does the chatbot stay on topic and answer the question asked?

(7) Application platform: not limited to Windows, Ubuntu, Linux, Android, Web browser, …etc.

(8) In a team size of 5 or 6 members.

   Please email team member details for consolidation by **20th September 2026** to:

   cheekwang.quah@singaporetech.edu.sg and if_jonathan.lim@singaporetech.edu.sg

Submission for Assessment

(1) Final technical report.

   Typical technical report would comprise:

   (i) Introduction/background/objectives

   (ii) Literature review and state-of-the-art

   (iii) System overview and architecture

   (iv) Method/approach

(v) Result, analysis and discussion

(vi) Conclusion

A report template in IEEE format is provided

Due: **Before week 13 (22nd November 2026)**.

(2) Package software

A compressed (**zipped format**) file containing all the source files and software libraries (including open-source libraries) for development of the project. Please name the file as: RSE2203_TeamXY.zip (e.g. RSE2203_Team01.zip)

Due: **before Week 13 (22nd November 2026).** Submit to **X-Site dropbox**.

(3) Presentation and Demo.

Video of presentation and demo of project (15 minutes).

Due: **before Week 13 (22nd November 2026)**. Submit to **X-Site dropbox**.

(4) An individual peer-review assessment form.

Due: **before Week 13 (22nd November 2026)**. Submit to **X-Site dropbox**.

Grading

(1) Presentation & demo (15%)
(2) Report (30%)
(3) Project quality and implementation (35%)
(4) Experiment, analysis and insight discussion (20%)

Suggestions for Potential Projects

(1) Robotics

Path-planning, motion planning, navigation, searching, perception and decision-making, automation, …etc. And e.g. ROS packages or nodes.

(2) Data analytics and visualization

Traffic/transport system, finance, learning and education, … etc.

(3) Artificial intelligence

Machine learning, decision tree, data modelling, …etc.

(4) Route planning and scheduling

Supply chain, logistic, production line, …etc.

(5) Foundation class/library

Software development kit, re-usable library, API, …etc.

(6) Others: recommender system, database system, signal processing, …etc.